

Proposal for Orbital 26

Team Name:

FoodCart

Proposed Level of Achievement:

Gemini

Motivation

Very often, customers visiting a restaurant can get bombarded with too many choices of food items on a menu, leading to decision paralysis. This leads to them taking a large amount of time to order, all while their stomachs' complaints grow louder. We want to create a smart ordering system that can recommend to customers what to eat, such that food can arrive at their table much faster.

Aim

We hope to create a food ordering system that recommends to customers what to order based on their budget, preferences and other metrics. The system should have all the essential features of a menu and ordering system, with the additional feature of food recommendations and order tracking.

We propose an intuitive swipe-based feature that allows users to quickly decide whether they are interested in a particular dish by swiping yes or no. Customers can either select a dish immediately to place an order or pick several options they would like to try. The system will also include a random picker that helps users decide among their selected dishes. Alternatively, users may allow the system to decide the final order automatically, without revealing the selected dish until the food is served.

The system will also include an order progress tracker, allowing customers to monitor the status of their orders and updates on the preparation and serving progress.

User Stories

1. As a customer who is *hungry*, I want to be able to decide on what to eat from quickly and keep track of the status of my order being prepared.
2. As a restaurant operator, I want to be able to manage food items and orders easily and seamlessly.

Features

	Customer	Admin
Core Features	• View all menu items • Filter menu items by category • View item details • Add items to cart • Update item quantity in cart • Place order • View order confirmation • Receive email confirmation for placed orders • Register / log in option	• Log in • Add menu items • Edit menu items • Delete menu items • Mark items as unavailable • View customer orders • Change order status: – Pending – Preparing – Ready – Completed
Unique Feature	1. Order Progress Tracking • Received • Preparing • Ready Soon • Served 2. Predicted Waiting Time • Each menu item has a preparation time used to estimate waiting time.	
Additional Features	1. Item Recommendations based on ordering patterns 2. Surprise Feature – integrate AI to randomly generate an order based on the user’s budget, preferences, and allergy information 3. Loyalty Program with a point system 4. Customer Dashboard showing past orders and most frequently ordered items 5. Real-time queue and estimated waiting time	1. Sales Analytics Dashboard • Top-selling items • Revenue today • Orders per hour 2. Photo Upload Feature – take a photo directly and automatically remove the background to add a new menu item 3. Restaurant Information Page showing restaurant details and opening hours
User Journey	Home page → Menu page → Item cards → Add to cart → Cart page → Checkout	Admin login → Admin dashboard → Manage menu

	form → Place order → Confirmation page	/ View orders → Update order status
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Feature	Timeline
Admin menu management	Early June – Mid June
Customer menu browsing	Early June – Mid June
Cart system	Early June – Mid June
Checkout and ordering	Early June – Mid June
Admin order management	Early June – Mid June
Unique features • Order progress tracking • Predicted waiting time	Mid June – End June
Customer account features	Mid June – End June
Additional features • Suggest items based on ordering patterns • Loyalty program and point system • Restaurant information page and opening hours • Sales analytics dashboard • Photo upload with automatic background removal for menu items • AI “surprise order” feature based on user budget, preferences, and allergies	Early July – End July

Timeline

Milestone	Date	Task	Details
1	Early May	Idea brainstorming, ideation, and research on existing products	Source and analyse existing products to prepare a proposal with full implementation details.
2	Mid May	Complete proposal	• Complete proposal with intended features and implementation details. • Start creating UI wireframes using Figma.
3	Mid May – Early June	Preparation and setup	• Gain hands-on experience with the selected tech stack and research unfamiliar technologies. • Set up Git repository and create branches. • Decide task allocation. • Prototype system interfaces before development begins. • Set up the initial project structure (skeleton directories and classes).
4	Early June	Proposal submission	Submit proposal for evaluation (1 June).
5	Early June – Mid June	System development (Phase 1)	• Begin building the system. • One team member focuses on backend development. • One team member focuses on frontend development. • Discuss challenges and difficulties encountered. • Core features with basic logic implemented.
6	Mid June	System integration and feature update	• Update progress on assigned tasks. • Integrate frontend and backend for a minimal working system with basic functionality. • Implement email confirmation and refine core features.
7	Mid June – End June	System extension and testing	• Extend system functionality. • Test features and refine system logic. • Implement unique features and customer account features.
8	End June	Prototype submission	Submit a prototype: a working system with core features implemented for evaluation (29 June).
9	Early July – Mid July	Feature expansion	• Improve and refine the existing system. • Start implementing additional features.
10	Mid July	Progress review and debugging	• Update progress on assigned tasks. • Cross-check functionality and debug issues. • Ensure all core features work as intended and refine implementations if necessary.

			• Develop additional features.
11	End July	Extended system submission	Submit an extended system: a working system with both core and extension features for evaluation (27 July).
12	Early August	Presentation preparation	• Start designing the project poster. • Record the demo video.
13	Mid August	Final deliverables	Complete and finalise the project poster.

Tech Stack

Web based application

- UI Wireframe: Figma
- Frontend: React
- Backend: Node.js + Express
- Database: MySQL / PostgreSQL | Firebase / Supabase
- Styling: Tailwind CSS
- Testing API: Postman
- Version control: Git + GitHub
- Poster: Canva + Figma

Qualifications

[Please elaborate on your technical skills and project experience of your team. This is required only for teams targeting Apollo 11 or Artemis.]

Software Engineering

[Please showcase the software engineering knowledge of your team by providing specific details about what software engineering related techniques / practices you plan to apply in the development process of your project. This is required only for teams targeting Apollo 11 or Artemis.]